Design of electrical power supply system in a typical petroleum refinery

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***Abstract*- The electrical system shall be designed economically for continuous and reliable services, safely to personnel and equipment, ease of maintenance and operation, minimum power losses, protection of equipment mechanically, interchangeability of equipment and addition of the of the loads. In order to achieve the above goals and obtain the desired results, a scientific study based on different theories and practical experiences will be needed.**

In this study, the power supply of Transnet Pipeline Terminal 2 (TM2) petroleum refinery in Heidelberg, the criteria and methods of designs of normal networks, electrical equipment and protection of the system will be discussed and investigated. A single line diagram will be presented as the outcome of the design. The single line diagram includes 88kV, 11kV and 400V voltage levels. In the second phase of the design single line diagram is consequently simulated by the ETAP software. The study will cover the following: load flow, short circuit current and motor starting.

The intention of this study is to create solution in different ways, electrical loads should be categorized in this energy industry as well as energizing these loads by a stable power supplies. In addition the key role of the short circuit impedance of the transformers in control of the short circuit will be presented. Furthermore the selection procedure of the electrical equipment and accessories including cables, transformers, circuit breaker, relays, and etc. are presented. The following factors such as the size of equipment, losses and voltage drops will be checked by load flow study. A comprehensive study of the short circuit current calculations is implanted and can be observed how the system can be checked by the results of this study. In the dynamic study of the system, the biggest motor starting is simulated and the impacts of the voltage dips due to starting of this motor on the other running motors are shown.

Since numerous types of equipment on one hand and the research on the economic matters on the other hand are time consuming, the scope of this report will mainly concentrate on the technical factors and as a results it does not cover the economic aspects. Moreover, high standard engineering in the petroleum industry is essential to design electrical system. It is noted that more economical options are acceptable as long as they end up with the same technical results or better.

Keywords: ***Power Systems, Petroleum, Distribution Networks, Electrical systems in hazardous areas, Relay selection, Circuit Breaker selection, Short circuit calculations, Load flow, motor starting.***

I. Introduction

In this paper the design of power supply in Transnet Pipeline New Multi Product Pipeline project (NMPP) Terminal 2 project is investigated. The project is done in two mainstages: basic design and detail design, in the basic design stage the location and process of refinery is studied and a rough estimate of load type and demand is consequently obtained. While in the detail design different kinds of networks and supplies (based on technical and economic situation) are surveyed and the best choice is selected and basic calculations, drawings and specifications are consequently provided.

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In the detailed design, precise drawings are considered a preference to purchase material and executing the project are issued.

a. Objectives

- To obtain deep understanding of electrical systems in the above mentioned industry.
- To know how to design a stable power system in the different projects by using a relevant software.
- To be able to analyze new power system in case of any possible problems and capability of finding the issues and solving them (trouble shooting).
- To obtain an ability of predicting the possible problems that may happen in power system.

b. Scope of project

Having a stable network in this industry is crucially important and power outage during operation could cost lots of money and time. So, an electrical expenditure is considered with little or no value when it comes in comparison to total above mentioned huge costs. Therefore, it is worth it spending time and energy during the design of stable networks to avoid any possible costly failure in the future.

In this report the electrical system of one unit of a refinery with two 88kV feeders and two main voltage levels of 11kV and 0.4kV have been studied. Although in descriptive parts it has been tried to illustrate the subject with a general discussion about other voltage levels, but in calculative sections only the above mentioned voltage levels have merely been considered based on work scope of the Company.

The main purpose of this thesis is to design a power supply with the right selection of electrical equipments. Therefore, other topics such as grounding, battery charger and UPS have not been discussed in details.

II. Methodology

In this project, the refinery power supply is simulated by ETAP software. In addition some electrical standard are the design criteria. Some big companies' manual such as ABB Ltd. is used as references.

To have a better overall understanding on how an electrical power supply system in a plant should be designed, the ABB Electrical power supply procedure is shown in the following page; [10]

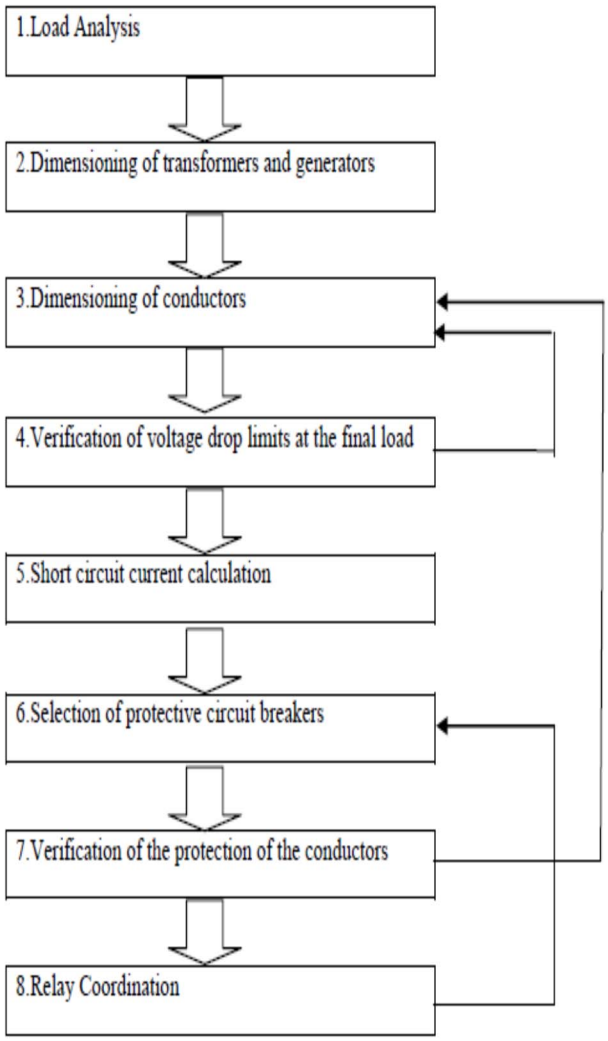


Figure 2.1: Design of Power Supply in a Plant

b. **Load analysis**:

Definition of power absorbed by the load and relevant position

Definition of the position of power centers (switchboards)

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Definition of the path and calculation of the length of connection elements

Definition of the total absorbed power, taking into accounts the utilization factors and demand factors

c. Dimensioning of transformers and generators:

15 till 30 percent margin should be considered for future

d. Dimensioning of conductors**:**

Evaluation of the current passing through conductors

Definition of the conductor type and insulation material

Definition of the cross section and the current carrying capacity

Calculation of the voltage drop at the load current in normal and transient (motor starting...) operation

e. Verification of the voltage drop limits at the final load

If the voltage drop is not in the limit, stage 3 should be modified

f. Short circuit current calculation

Maximum value at the bus bar and minimum value at the end of the line

g. Selection of protective circuit breakers with:

Breaking capacity higher than the maximum prospective short circuit current

Rated current no lower than load current

Characteristics compatible with the type of protected load (motors, capacitors...)

h. Verification of the protection of the conductors

Verification of the protection against over load: The rated current or the set current of the circuit breaker shall be higher than the load current but lower than the current capacity of the conductor

Verification of the protection against short circuit: The specific load through energy by the circuit breaker under short circuit condition shall be lower than the specific energy let through energy which can be withstood by the cable ($I^2t \leq k^2S^2$)

In case of obtaining negative outcome, all the above stages shall be repeated from stage 3

i. Verification of the coordination with other equipment (Relay coordination)

In case of obtaining negative outcome, all the above stages shall be repeated from stage 6

III. Design criteria

In order to be certain that our design has acquired technical quality, the engineering part of the job must be done according to the standards and technical specifications. Hereunder, some important criteria are presented.

j. Voltage level

The following voltage levels have been selected for electrical system at the rated frequency of 50 HZ. Equipment will be suitable for continuous operation with voltages variation within 5% of nominal values.

k. Transformer

In case of trip on one transformer, another transformer should be able to withstand the entire downstream load. In addition each transformer should at least have 20% spare in normal operation according to IPS-E-EL-100. [8]

The short-circuit voltage in percent V_K% according IEC 60076-5 for 2.5 MVA and 12.5 MVA should be 6% and 8% respectively.[4]

l. Preparation of load list

This document shows the loads in each refinery unit. Usually required loads are specified by process department and supplementary information is completed by mechanical and then electrical departments. It should be noted that preliminary data are just estimated data and the precise data will be reached from vendors during project. These are objectives of issuance of this document:

Recognizing industrial and non-industrial loads of projects in order to provide single line diagrams, cable schedules, cable route plans….

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Control of loads variations during the design and construction of the project and updating the relevant data.

Specifying total loads of switchgears and MCC's and consequently their normal current.

Transformer sizing. Calculation of total loads of the project in order to generator sizing for local power plant or purchasing demand power from regional electric power company.

Calculation of the emergency loads and method of providing (emergency generator rating calculations)

Determination of the maximum load of the project in order to specifying power demand.

***Feeding types*:**

Different methods of feeding for different loads can be considered as follows;

- Normal feeding (N):
- These loads are fed from just normal buses and incase of any fault the loads will lose their power supply.
- Normal feeding with reacceleration (NR)
- These kinds of loads are fed from normal bus bars too, but in case of short interruption power (usually under voltage) they are capable of restarting so fast.
- Essential feeding:
- The feed is assured by an emergency power generator.
- Vital feeding:
- It implies that no interruption in the power supply is allowed.
- ***Load types and required power***
- Motor loads
- Heaters
- Lighting & Socket loads
- UPS & DC loads

m. Cable sizing

For determination of electrical power cables it is necessary to do the following studies:

- Cable Ampacity
- Short circuit withstanding current
- Voltage drop

Since there are different methods for physical arrangement of cables as well as a possibility of having different environmental and physical conditions, therefore before cable sizing, it is necessary to accurately consider the physical and environmental condition of cable route.

IV. Preliminary single line diagram

After preparing load list(s) and understanding the demand, we need to design a stable network to supply our required power. During the design, feeding type mentioned in the load list and voltage level should be taken into account so that we would get general idea how to feed the loads. Normal loads are fed from the normal bus bar and essential loads such as emergency lightings are fed from the emergency bus bar.

Vital loads such as Emergency shutdown (ESD) systems are fed from UPS.

After conducting our detailed calculation, the final single line diagram would be different from the primary one as shown in the figure 3.1. Consequently, when the final single line is shown, the modification process can be found.

As it is shown in the following, this unit of the refinery is supplied by two 20kV incomings and the 12.5 MVA transformers provide the 6.3kV output voltage. They must work as a backup of each other so that one transformer can create enough capacity to feed all the loads in case of any possible failure in another one. Of what was discussed in the above, it can be concluded that each transformer in the normal function is loaded by half of the total load.

Although each of these two normal transformers are able to feed all the loads in the normal function, however, another 20kV line has also been considered for the emergency load purposes in order to ascertain the availability of power supply during any faulty condition in the system. On the other hand, "automatic change over system" has been designed for the purpose of feeding the emergency loads in case of missing the normal bus bars.

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The vital loads are fed by UPS and the source of these loads is the batteries which are charged during normal operation. Moreover, emergency operation is also used in case of having none of the normal or emergency supplies.

The sizing of transformers, bus bars, and circuit breakers should be performed by load balance study as well as software simulations which is mentioned in this and next chapters.

V. Conclusion

As mentioned in the previous chapters, the purpose of this study is to design the stable electrical system in refineries. Considering the following items are important key measurements that ascertain a safe, stable and continuous power supply:
Precise load list in terms of required power, load factors, load type and relevant feeding type assures a reliable system. Since the main purpose of electrical network is to energize electrical loads, the better understanding of loads, the better design of the network.

Taking normal feeding, emergency feeding and vital feeding (for relevant loads) into consideration are crucially important.

Reliable protections including relays and circuit breakers to protect equipment and personnel in faulty situation must be paid attention to.

In addition, right connection of electrical protection and relays with other instrumentation systems such as fire & gas protection system, distribution control system, emergency shutdown system will assure safe refinery.
Main studies including load flow, short circuit and transient study assist us to confirm stability of system. Furthermore, it should be noted that any modifications to each study will influence on the results of other studies and in the final stages, entire studies' results should be checked in order to approve all desired results.

Cable sizing is important in many aspects such as cost, voltage drop and reactive power losses. Wrong cable sizing can adversely affect the system by leading to faulty conditions as well as huge costs of detecting the fault and replacing the cable which can be costly and time consuming.

Although in this project the power factor was acceptable according to design criteria, but capacitor bank is recommended for improving power factor for the purpose of having lower reactive losses, better control on voltage drops and lower size of cable. Selecting smaller cable cross section can be more economical.

By adjusting short circuit impedance of transformer, it is possible to control short circuit current. But the higher short circuit impedance, the higher losses. That is why utilizing transformers with higher sizes are sometimes recommended. Considering the tap changer for transformer can assure the required voltage level at site.

8. Right equipment selection for hazardous areas of refineries can avoid explosion and any possible disaster.

9. To ascertain the ability of the motors to start up the required torque must be provided and also start up time is an important factor that should be as short as possible.

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